

Diagnosis: Colorectal cancer

Treatment 1: Crizotinib

Chosen clinical trials:

	Title	Phase	Group	Population	CR	PR	ORR	SD	DCR	PFS	OS
1	MEK and MET Inhibition in Colorectal Cancer (NCT02510001)	completed phase1	Dose Expansion Phase	30	0.0	0.0	0.0	23.33	0.0	0.0	0.0
			Dose Expansion Phase	37	0.0	0.0	0.0	0.0	0.0	1.81	5.42

Chosen scientific papers:

Crizotinib, when combined with vemurafenib, showed rapid and marked effectiveness in treating BRAF-mutated colorectal cancer (clinical trial) [1]. Combination binimetinib/crizotinib showed poor tolerability with no objective responses observed in RASMT advanced CRC patients (clinical trial; 36 patients) [2]. Molecularly targeted treatment with crizotinib induced a rapid and sustained partial response, but disseminated tumor progression occurred after 15 months (clinical trial) [3]. The expression level of SDHB was correlated with drug sensitivity to Crizotinib (clinical trial) [4]. Crizotinib achieved a partial response (PR) with progression-free survival (PFS) of 3 months in an ALK-fusion mCRC patient (case report; 1 patient) [5]. Crizotinib delayed the progression of liver metastases and peritoneal carcinomatosis in colorectal cancer models (in vivo) [6]. Crizotinib combined with MMC showed synergistic anti-cancer effects in CRC cell lines and xenograft models (in-vivo CRC xenograft model) [7]. Crizotinib blocked the HGF/STAT3/SOX13/c-MET axis, significantly inhibiting SOX13-mediated CRC migration, invasion, and metastasis (clinical trial) [8]. Crizotinib, when combined with cetuximab or panitumumab, significantly reduces colony number and inhibits tumor growth in colorectal cancer (in vitro and in vivo) [9]. Crizotinib reduced metastasis in all models tested when used as a single agent or in combination with sunitinib (in vivo) [10].

Literature:

[1] MET-Driven Resistance to Dual EGFR and BRAF Blockade May Be Overcome by Switching from EGFR to MET Inhibition in BRAF-Mutated Colorectal Cancer. <https://pubmed.ncbi.nlm.nih.gov/27325282>

[2] A Phase Ia/b study of MEK1/2 inhibitor binimetinib with MET inhibitor crizotinib in patients with RAS mutant advanced colorectal cancer (MErCuRIC). <https://pubmed.ncbi.nlm.nih.gov/40211189>

[3] ROS1 genomic rearrangements are rare actionable drivers in microsatellite stable colorectal cancer. <https://pubmed.ncbi.nlm.nih.gov/36053834>

[4] TBX15 and SDHB expression changes in colorectal cancer serve as potential prognostic biomarkers. <https://pubmed.ncbi.nlm.nih.gov/38378070>

[5] Clinical Responses to Crizotinib, Alectinib, and Lorlatinib in a Metastatic Colorectal Carcinoma Patient With ALK Gene Rearrangement: A Case Report. <https://pubmed.ncbi.nlm.nih.gov/34036227>

[6] In vivo imaging xenograft models for the evaluation of anti-brain tumor efficacy of

targeted drugs. <https://pubmed.ncbi.nlm.nih.gov/29125233>

[7] Preclinical rationale for combination of crizotinib with mitomycin C for the treatment of advanced colorectal cancer. <https://pubmed.ncbi.nlm.nih.gov/28886275>

[8] SOX13 promotes colorectal cancer metastasis by transactivating SNAI2 and c-MET. <https://pubmed.ncbi.nlm.nih.gov/32111984>

[9] Broad-spectrum receptor tyrosine kinase inhibitors overcome <https://pubmed.ncbi.nlm.nih.gov/30863492>

[10] HGF/c-Met pathway is one of the mediators of sunitinib-induced tumor cell type-dependent metastasis. <https://pubmed.ncbi.nlm.nih.gov/22269210>